

ARCOS Group

uc3m | Universidad **Carlos III** de Madrid

Introduction to the course

Computer Structure

Bachelor in Computer Science and Engineering
Bachelor in Applied Mathematics and Computing
Dual Bachelor in Computer Science and Engineering and Business Administration



Introduction



- ▶ **General information about the course**
- ▶ Course development
- ▶ Evaluation system

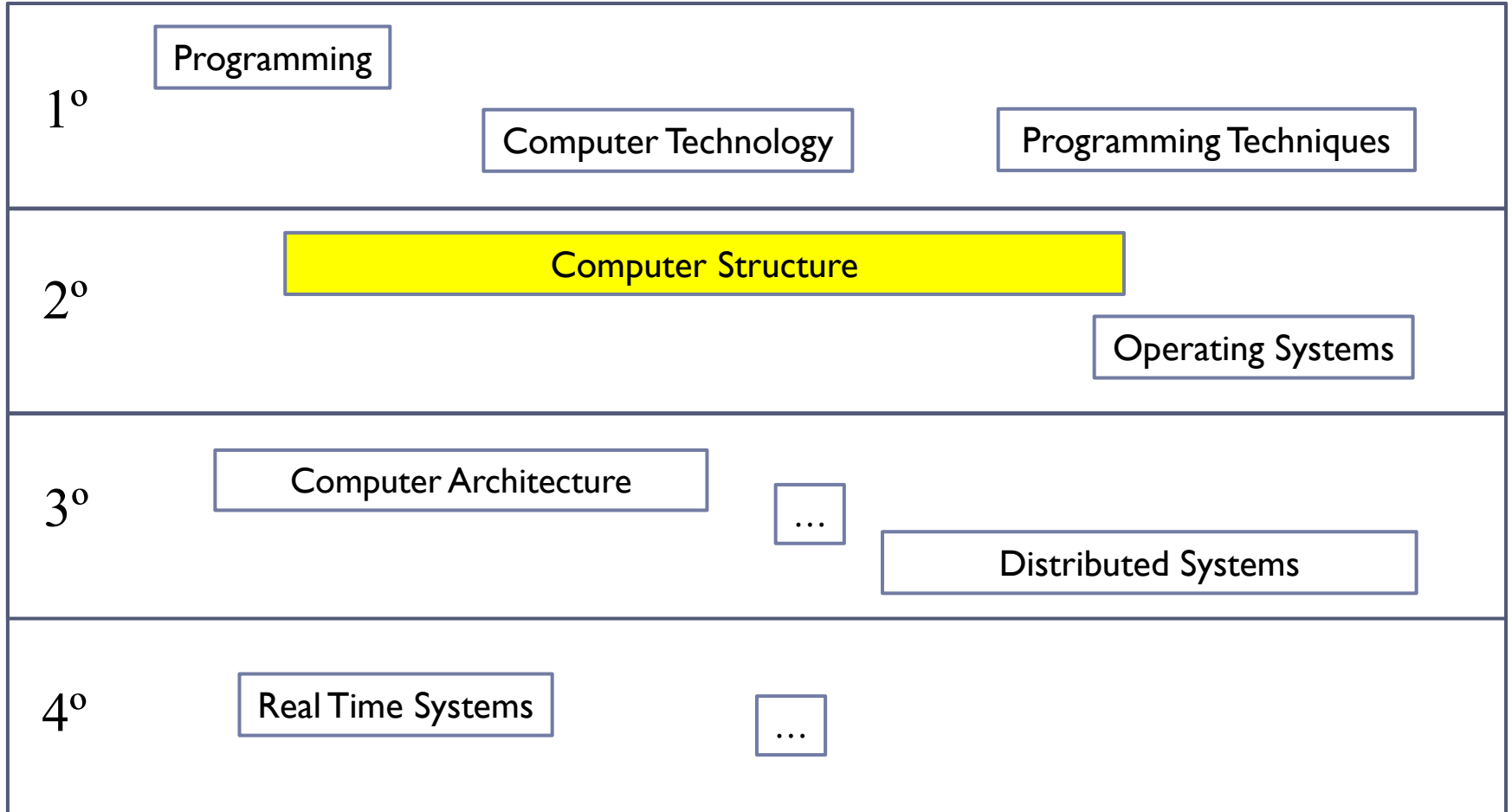
Computer Structure at UC3M

- ▶ YEAR: 2º
- ▶ QUARTER: 1º
- ▶ ECTS credits: 6
- ▶ MANDATORY / BASIC INSTRUCTION

- ▶ Grades:
 - ▶ Bachelor in Computer Science and Engineering
 - ▶ Bachelor in Applied Mathematics and Computing
 - ▶ Dual Bachelor in Computer Science and Engineering and Business Administration

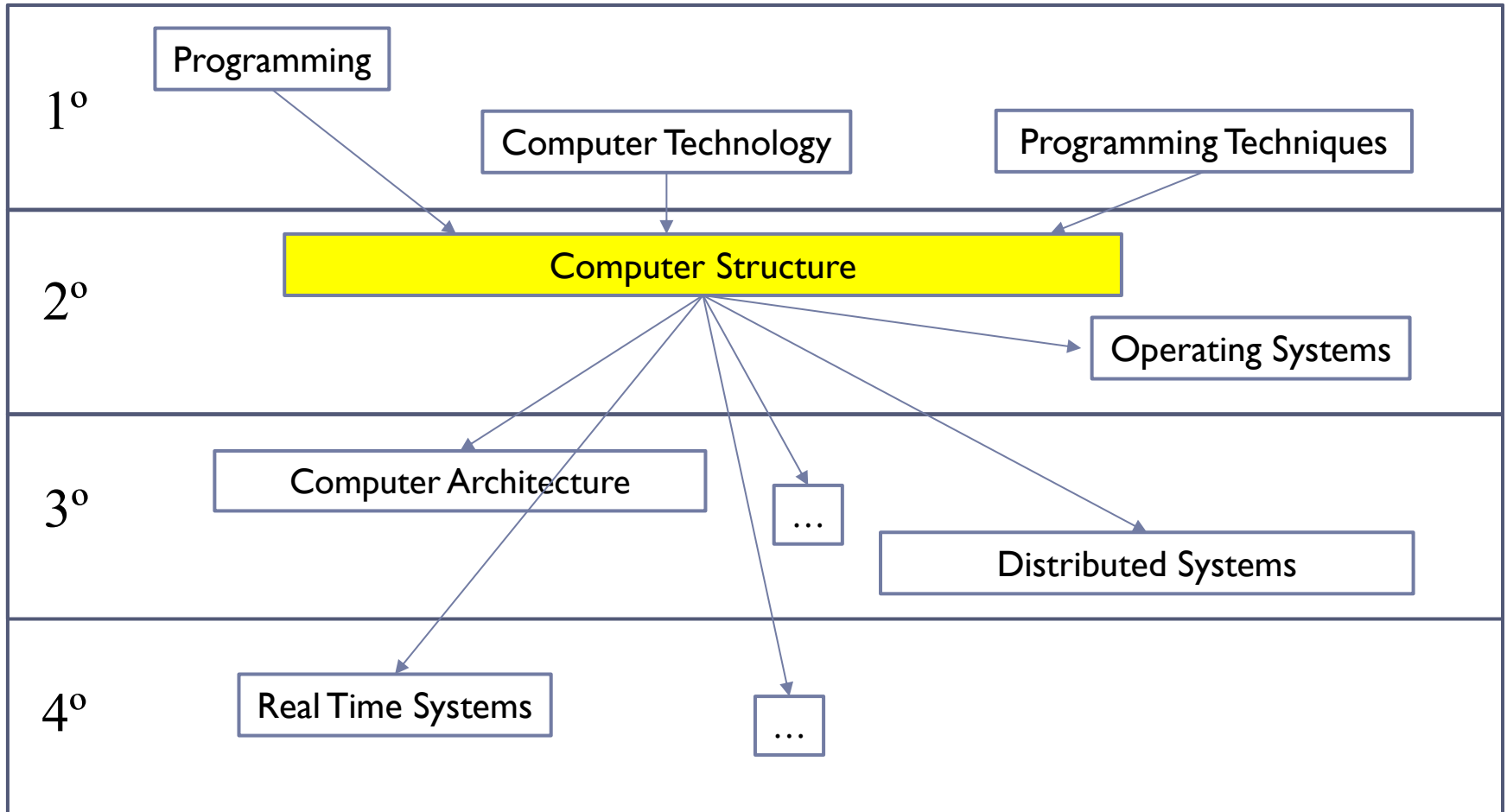
Computer Structure

Bachelor in Computer Science and Engineering



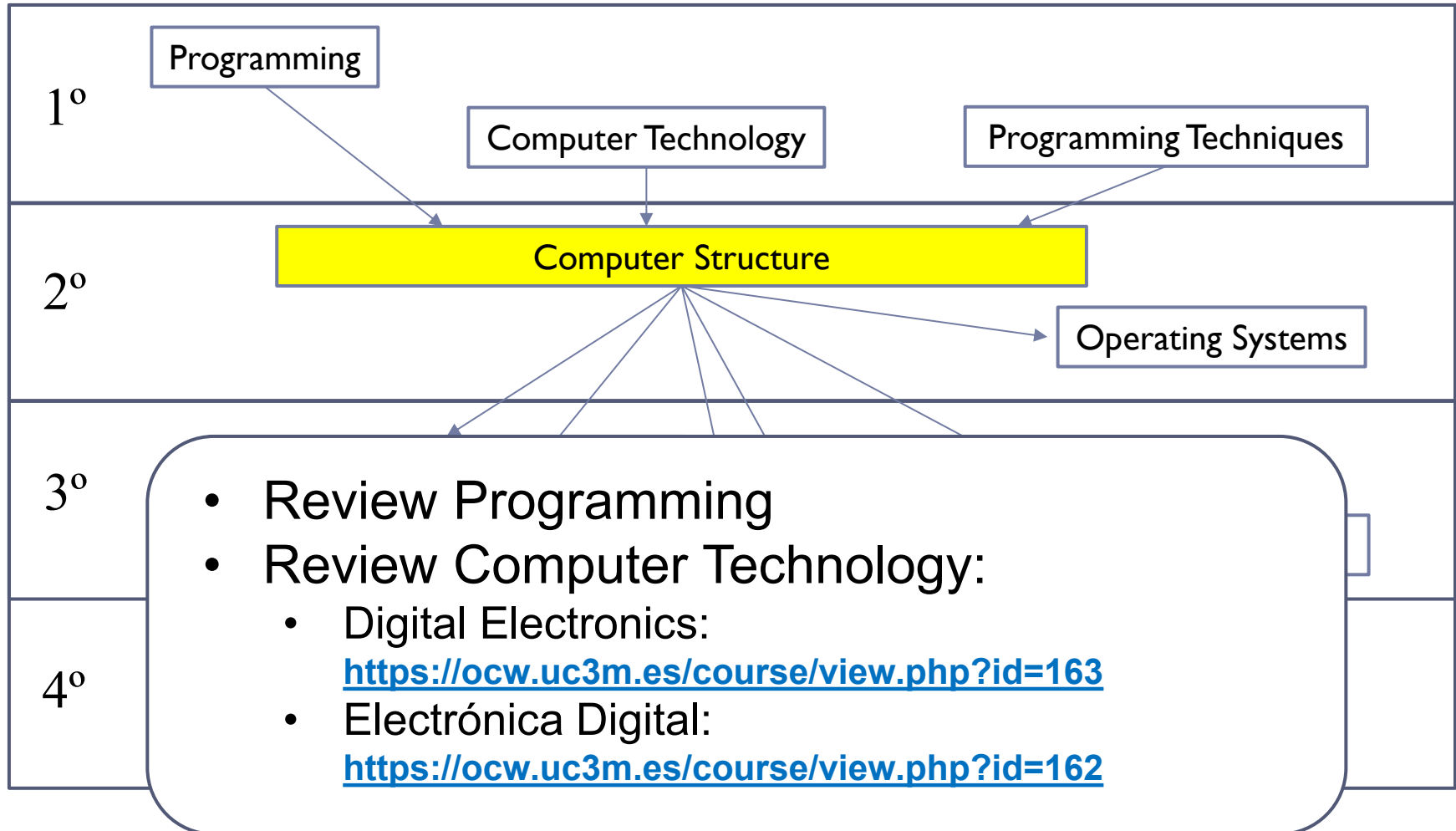
Computer Structure

Bachelor in Computer Science and Engineering

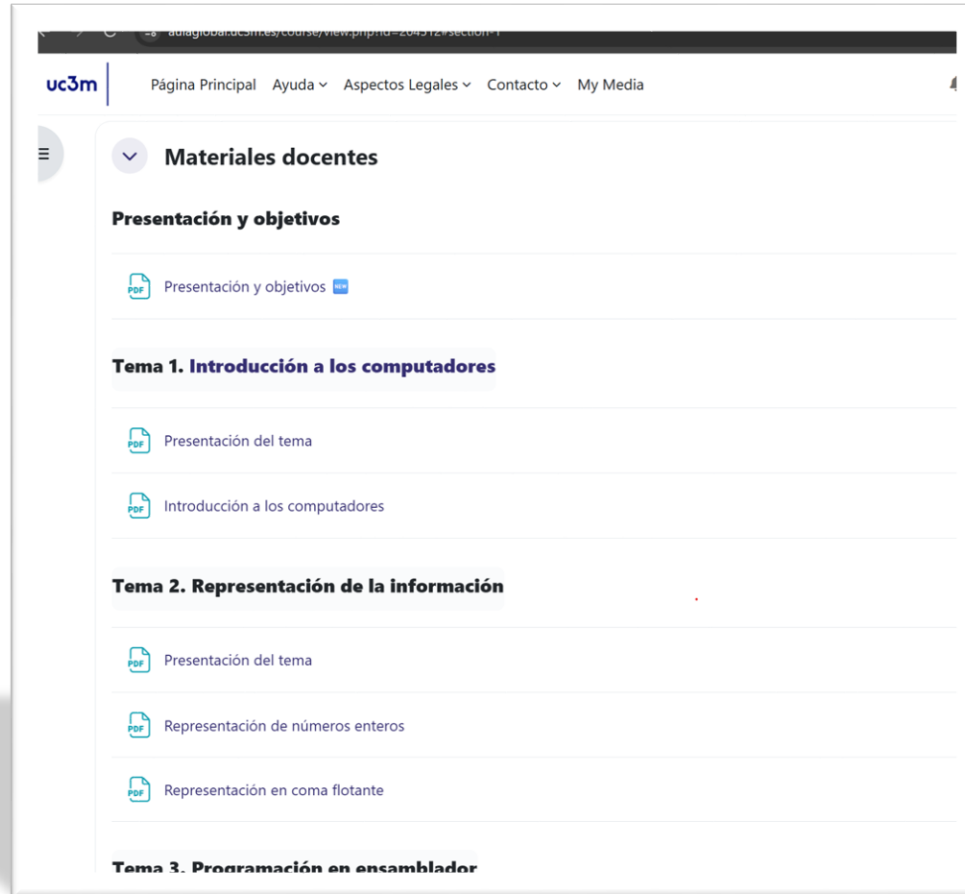


Computer Structure

Bachelor in Computer Science and Engineering



Primary source of information



Aula Global

Summary course profile

- ▶ COMPUTER STRUCTURE

- ▶ **Coordinator**

- ▶ **Goals**

- ▶ **Program**

- ▶ **Bibliography**

Course profile

Coordinator

- ▶ **COMPUTER STRUCTURE**
- ▶ General coordinator: **Félix García Carballeira**
(felix.garcia@uc3m.es)
- ▶ Bachelor in Computer Science and Engineering
 - ▶ Félix García Carballeira (felix.garcia@uc3m.es)
- ▶ Bachelor in Applied Mathematics and Computing
 - ▶ Dante Domizzi Sánchez Gallegos (dantsanc@pa.uc3m.es)
- ▶ Dual Bachelor in Computer Science and Engineering and Business Administration
 - ▶ Alejandro Calderón Mateos (acaldero@inf.uc3m.es)

Course profile

Coordinator

- **Report any issues with the course's progress as soon as possible**

- ▶ General coordinator: [Félix García Carballeira](mailto:felix.garcia@uc3m.es)
(felix.garcia@uc3m.es)
- ▶ Bachelor in Computer Science and Engineering
 - ▶ Félix García Carballeira (felix.garcia@uc3m.es)
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Course profile

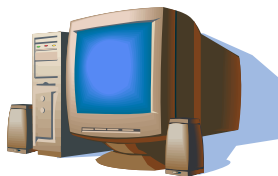
Goals

- ▶ COMPUTER STRUCTURE
- ▶ To know and understand the main components and basic operation of a computer (concepts behind the design of computers).

Course profile

Goals

- ▶ COMPUTER STRUCTURE
- ▶ To **know** and **understand** the **main components** and **basic operation** of a computer (concepts behind the design of computers).



Course profile

Goals

- ▶ COMPUTER STRUCTURE
- ▶ To know and understand the main components and basic operation of a computer (concepts behind the design of computers).

High-level language

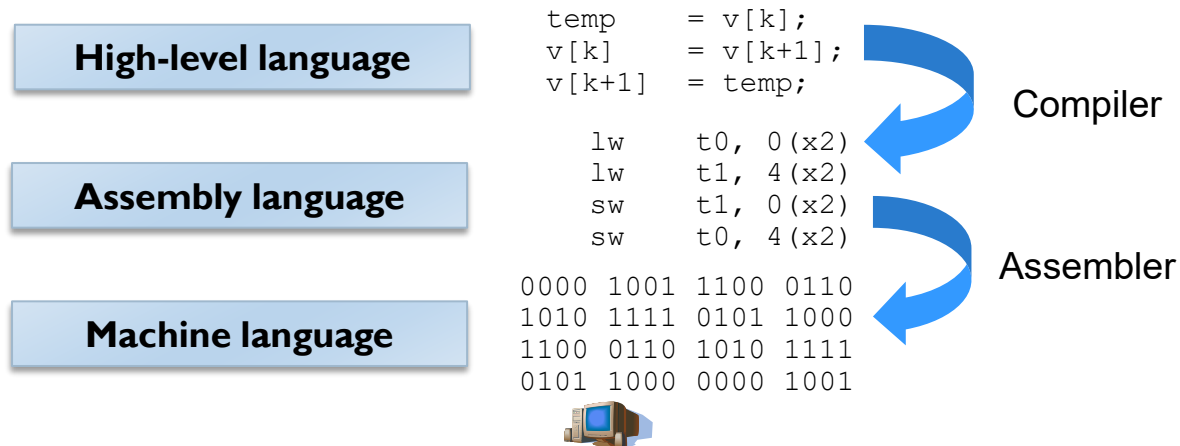
```
temp    = v[k];  
v[k]    = v[k+1];  
v[k+1]  = temp;
```



Course profile

Goals

- ▶ **COMPUTER STRUCTURE**
- ▶ To **know** and **understand** the **main components** and **basic operation** of a computer (concepts behind the design of computers).

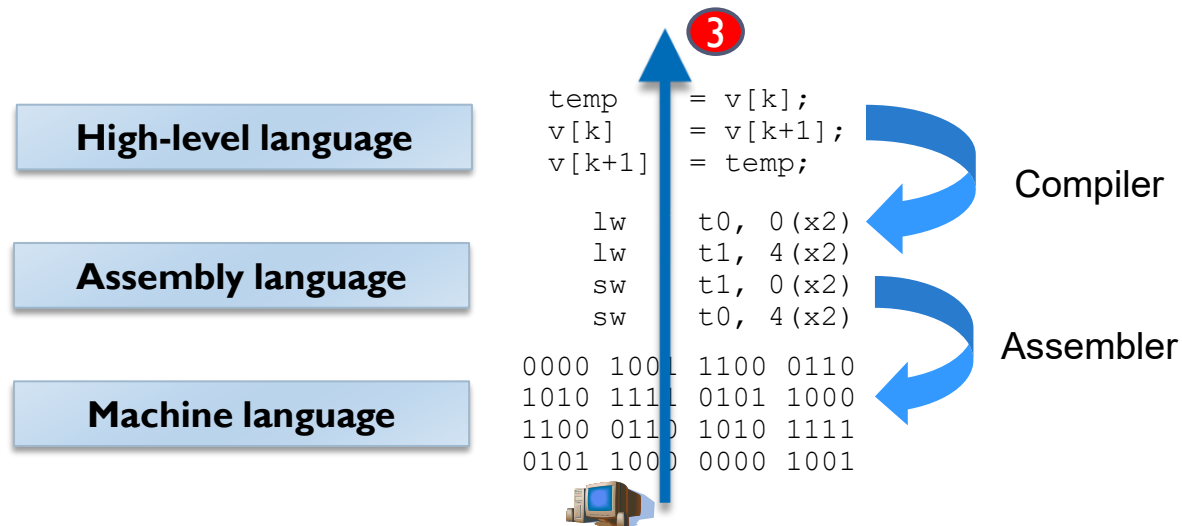


Course profile

Goals

► COMPUTER STRUCTURE

- To **know** and **understand** the **main components** and **basic operation** of a computer (concepts behind the design of computers).



Course profile

Goals

- ▶ COMPUTER STRUCTURE
- ▶ To know and understand the main components and basic operation of a computer (concepts behind the design of computers).

Ejemplo 1

- ▶ Sea el siguiente código:

```
int n;  
n = 40000;  
printf("%d \n", n*n ); // 40000²  
  
n = 50000;  
printf("%d \n", n*n ); // 50000²
```

- ▶ Produce la siguiente salida:

```
1600000000  
-1794967296
```

- ▶ ¿Es correcto?, ¿Cuál es el problema?

Ejemplo 2

- ▶ Sea el siguiente código:

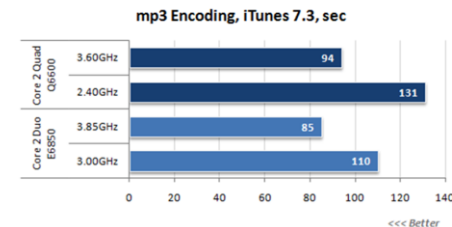
```
float x, y, z;  
  
x = 1.0e20; y = -1.0e20; z = 3.14;  
  
printf("%f\n", (x + y) + z);  
printf("%f\n", x + (y + z));
```

- ▶ Produce la siguiente salida:

```
3.140000  
0.000000
```

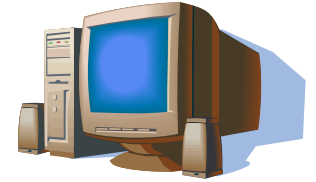
- ▶ ¿Se cumple $(x+y) + z == x + (y+z)$?

Ejemplo 3



- ▶ ¿Es más rápido un procesador con dos núcleos o un procesador con cuatro núcleos?

Example 1



- ▶ Let the following code be:

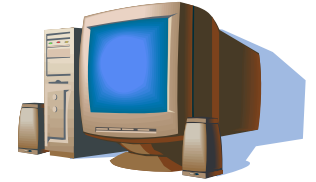
```
int n;  
n = 40000;  
printf("%d \n", n*n ); // 400002  
  
n = 50000;  
printf("%d \n", n*n ); // 500002
```

- ▶ It produces the following output:

```
1600000000  
-1794967296
```

- ▶ Is this correct, what is the problem?

Example 2



- ▶ Let the following code be:

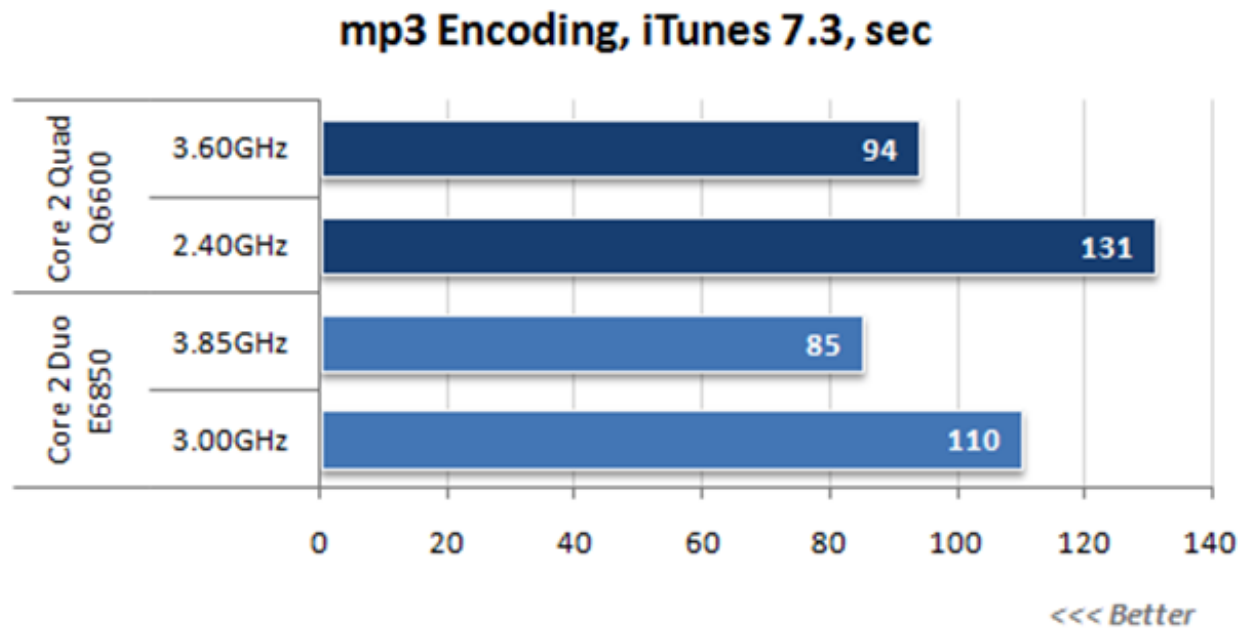
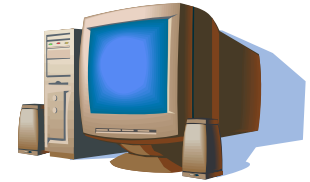
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printf("%f\n", x + (y + z));
```

- ▶ It produces the following output:

```
3.140000  
0.000000
```

- ▶ Is $(x+y) + z == x + (y+z)$ satisfied?

Example 3



- Is a dual-core processor faster than a quad-core processor?

Course profile

Goals

- ▶ COMPUTER STRUCTURE
- ▶ To know and understand the main components and basic operation of a computer (concepts behind the design of computers).

Ejemplo 1



- ▶ Sea el siguiente código:

```
int n;  
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printf("%d \n", n*n ); // 40000²  
  
n = 50000;  
printf("%d \n", n*n ); // 50000²
```

- ▶ Produce la siguiente salida:

```
1600000000  
-1794967296
```

- ▶ ¿Es correcto?, ¿Cuál es el problema?

Ejemplo 2



- ▶ Sea el siguiente código:

```
float x, y, z;  
  
x = 1.0e20; y = -1.0e20; z = 3.14;  
  
printf("%f\n", (x + y) + z);  
printf("%f\n", x + (y + z));
```

- ▶ Produce la siguiente salida:

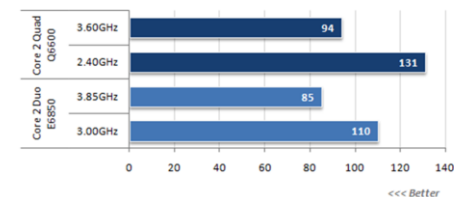
```
3.140000  
0.000000
```

- ▶ ¿Se cumple $(x+y) + z == x + (y+z)$?

Ejemplo 3



mp3 Encoding, iTunes 7.3, sec



- ▶ ¿Es más rápido un procesador con dos núcleos o un procesador con cuatro núcleos?

Course profile

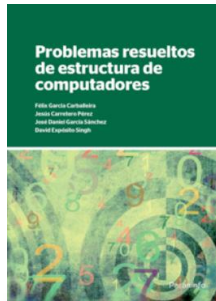
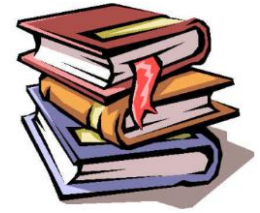
Program



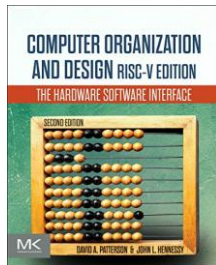
- ▶ Lesson 1. Introduction to computers
- ▶ Lesson 2. Data representation and basic
- ▶ Lesson 3. Introduction to assembly programming
- ▶ Lesson 4. Processor
- ▶ Lesson 5. Memory hierarchy
- ▶ Lesson 6. Input/output systems

Bibliography

basic



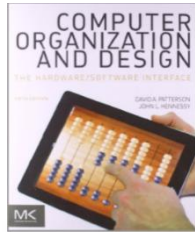
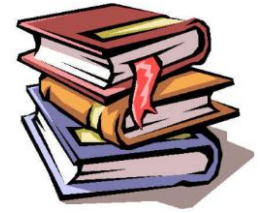
- **Problemas resueltos de Estructura de Computadores**
F. G. Carballeira, J. C. Pérez, J. D. García, D. E. Singh,
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Editorial Paraninfo, 2015



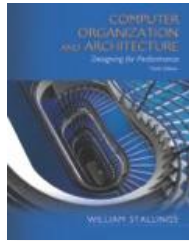
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David A. Patterson, J. L. Hennessy,
Second edition, 2021

Bibliography

complementary



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D.A. Patterson, J. Hennessy
5th edition,
Morgan Kaufmann Series, 2013



- ▶ **Computer Organization and Architecture.**
William Stallings
10th edition,
Editorial Pearson, 2016



- ▶ **Fundamentos de Sistemas Digitales.**
Thomas L. Floyd
11th edition
Editorial Pearson, 2016

Introduction



- ▶ General information about the course
- ▶ **Course development**
- ▶ Evaluation system

Course Scheduling

Week	Content	Labs.	Assignment submission	Midterm exam
1	Presentation, introduction, and review of information representation.			
2	Floating point and fundamentals of assembly programming.			
3	Branches and control structures in assembly, and exercises.	Lab. 1		
4	Memory access instructions, data representation, and floating-point instructions.			
5	Instruction formats and introduction to function calls in assembly.			
6	Function calls and exercises.			
7	Elements and organization of an elementary processor.	Lab. 2	Assignment 1	
8	Instruction execution.	Lab. 3		
9	Control unit and MIDTERM EXAM.			M. exam
10	Control unit, interrupt handling, and processor boot.			
11	Memory systems and introduction to cache memory.	Lab. 4		
12	Cache memory (cont.) and introduction to virtual memory.			
13	Virtual memory and exercises.		Assignment 2	
14	I/O (Input/Output)			

Schedule



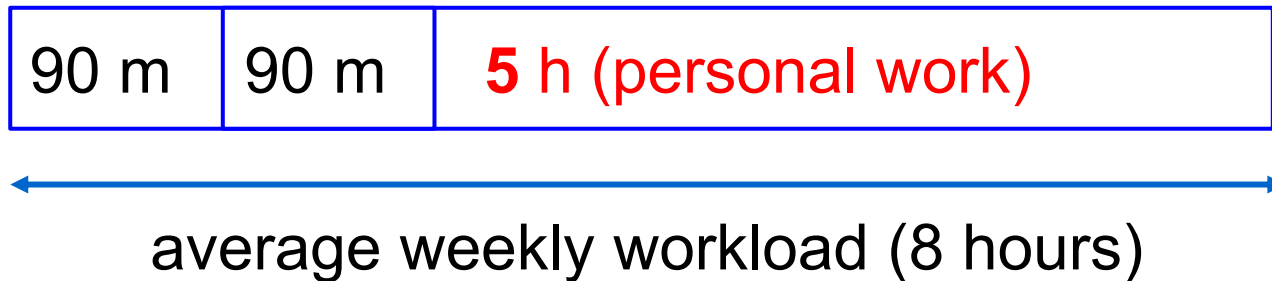
- ▶ 14 weeks in total with 32 sesiones of 90 minutes (presential classes)
- ▶ 10 weeks of 2 sessions per week
- ▶ 4 weeks of 3 sessions per week
- ▶ Additional laboratory session (presential)

Week	Content	Labs.	Assignment submission	Midterm exam
1	Presentation, introduction, and review of information representation.			
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Schedule



- ▶ 14 weeks in total with 32 sesiones of 90 minutes (presential classes)
 - ▶ 10 weeks of 2 sessions per week
 - ▶ 4 weeks of 3 sessions per week
 - ▶ Additional laboratory session (presential)



Teachers and classrooms

E.g.: Leganés, Bachelor on CS&E



uc3m | Horarios del curso 2026/2027

Horarios del curso 2026/2027

Escuela Politécnica Superior (Leganés): Doble Grado en Ingeniería Informática y Administración de Empresas

Fecha de generación: 2026-09-04 08:01:21

→ La información de los horarios de esta titulación ha sido generada de forma automática. Los cambios realizados durante esta jornada no se reflejarán instantáneamente.
→ La equivalencia de semanas para los horarios puede consultarla en la siguiente tabla (la fecha corresponde al lunes de la semana en cuestión).

2º Cuatrimestre	S1→ 28/01	S2→ 01/02	S3→ 08/02	S4→ 15/02	S5→ 22/02	S6→ 01/03	S7→ 08/03	S8→ 15/03	S9→ 22/03	S10→ 29/03	S11→ 05/04	S12→ 12/04	S13→ 19/04	S14→ 26/04	S15→ 03/05
	S16→ 10/05	S17→ 17/05	S18→ 24/05	S19→ 31/05	S20→ 07/06	S21→ 14/06	S22→ 21/06	S23→ 28/06							

1º Cuatrimestre	S1→ 27/07	S2→ 03/08	S3→ 10/08	S4→ 17/08	S5→ 24/08	S6→ 31/08	S7→ 07/09	S8→ 14/09	S9→ 21/09	S10→ 28/09	S11→ 05/10	S12→ 12/10	S13→ 19/10	S14→ 26/10	S15→ 02/11
	S16→ 09/11	S17→ 16/11	S18→ 23/11	S19→ 30/11	S20→ 07/12	S21→ 14/12	S22→ 21/12	S23→ 28/12	S24→ 04/01	S25→ 11/01	S26→ 18/01	S27→ 25/01			

Expandir todos los nodos del árbol Contraer todos los nodos del árbol

▼ HORARIOS DE LA TITULACIÓN POR CURSO

▼ Formación Obligatoria y Básica

▼ Formación Obligatoria

▶ 1º Curso

▼ 2º Curso

▼ 1º Cuatrimestre

▶ 13160 - Estadística II, 6 ECTS

▼ 13874 - Estructura de Computadores, 6 ECTS

▼ 🇪🇸 Grupo 81

Responsable: GARCIA CARBALLEIRA, FELIX			
Mie	10:45-12:15	Semanas: 7-20	Aulas: 4.1.E01 (Leganés)
Jue	12:30-14:00	Semanas: 7-20	Aulas: 2.3.C03 (Leganés)

▼ 🇪🇸 Grupo 82

Responsable: CALDERON MATEOS, ALEJANDRO			
Mie	15:30-17:00	Semanas: 7-20	Aulas: 2.3.A01 (Leganés)
Jue	17:15-18:45	Semanas: 7-20	Aulas: 2.3.D03 (Leganés)

Introduction

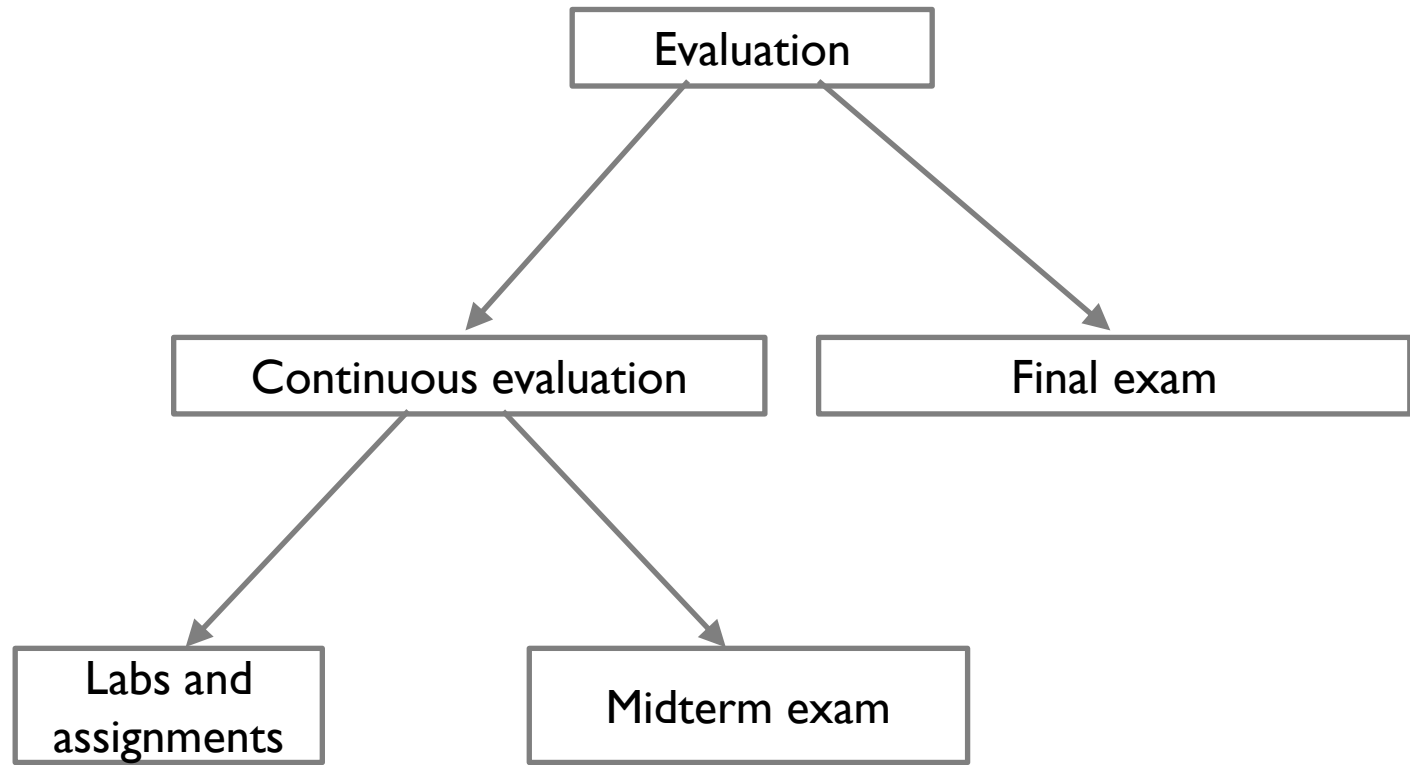


- ▶ General information about the course
- ▶ Course development
- ▶ **Evaluation system**

Evaluation



- The student evaluation will be based on:



Evaluation



1. Ordinary call

- ▶ Continuous evaluation is followed
- ▶ No continuous evaluation is followed

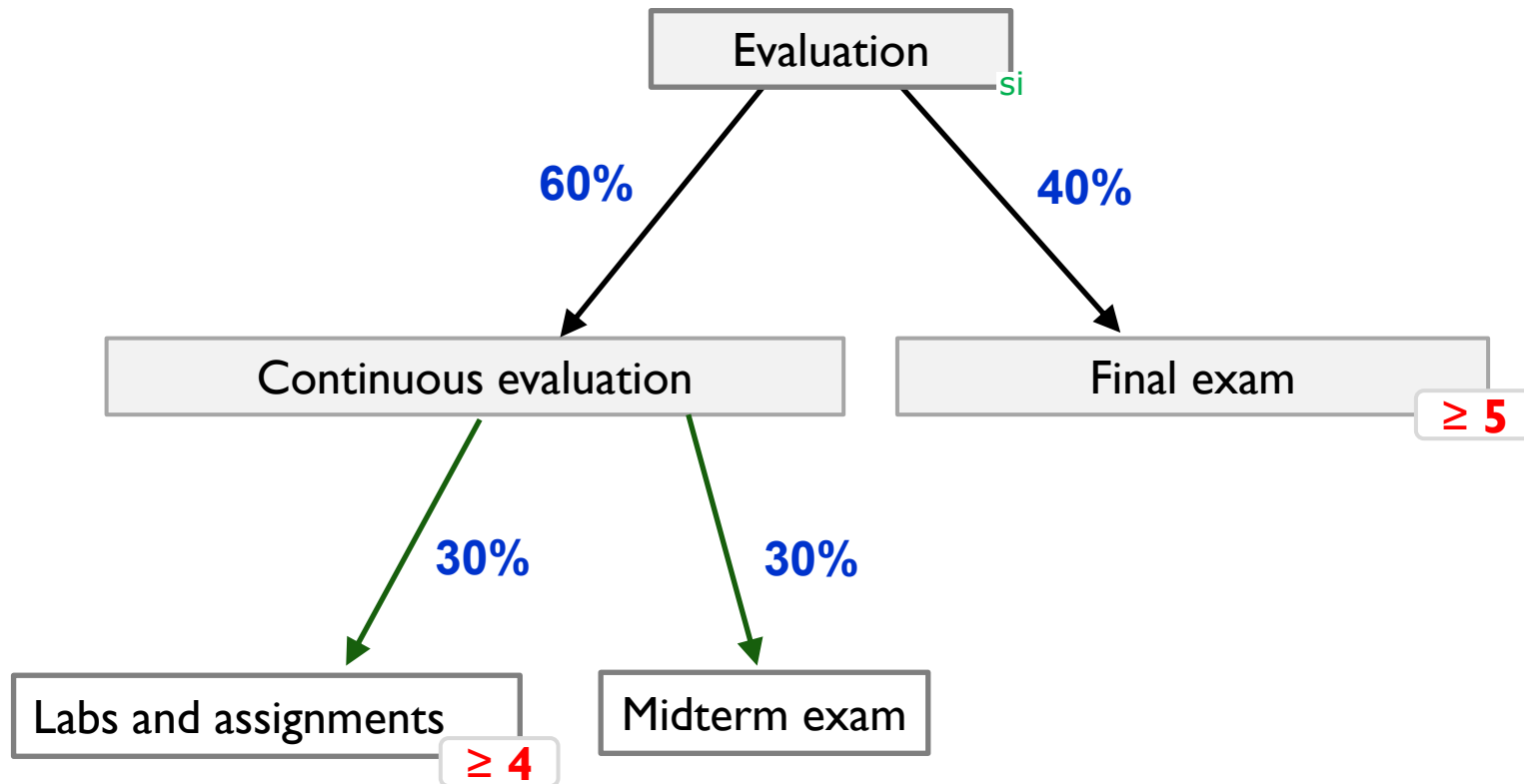
2. Extraordinary call

- ▶ Continuous evaluation is followed
- ▶ No continuous evaluation is followed

Evaluation

	Ordinary call	Extraordinary call
Continuous eval. is followed	Ordinary call w/ continuous eval.	extraordinary call w continuous eval.
Continuous eval. is NOT followed	Ordinary call w/o continuous eval.	extraordinary call w/o continuous eval.

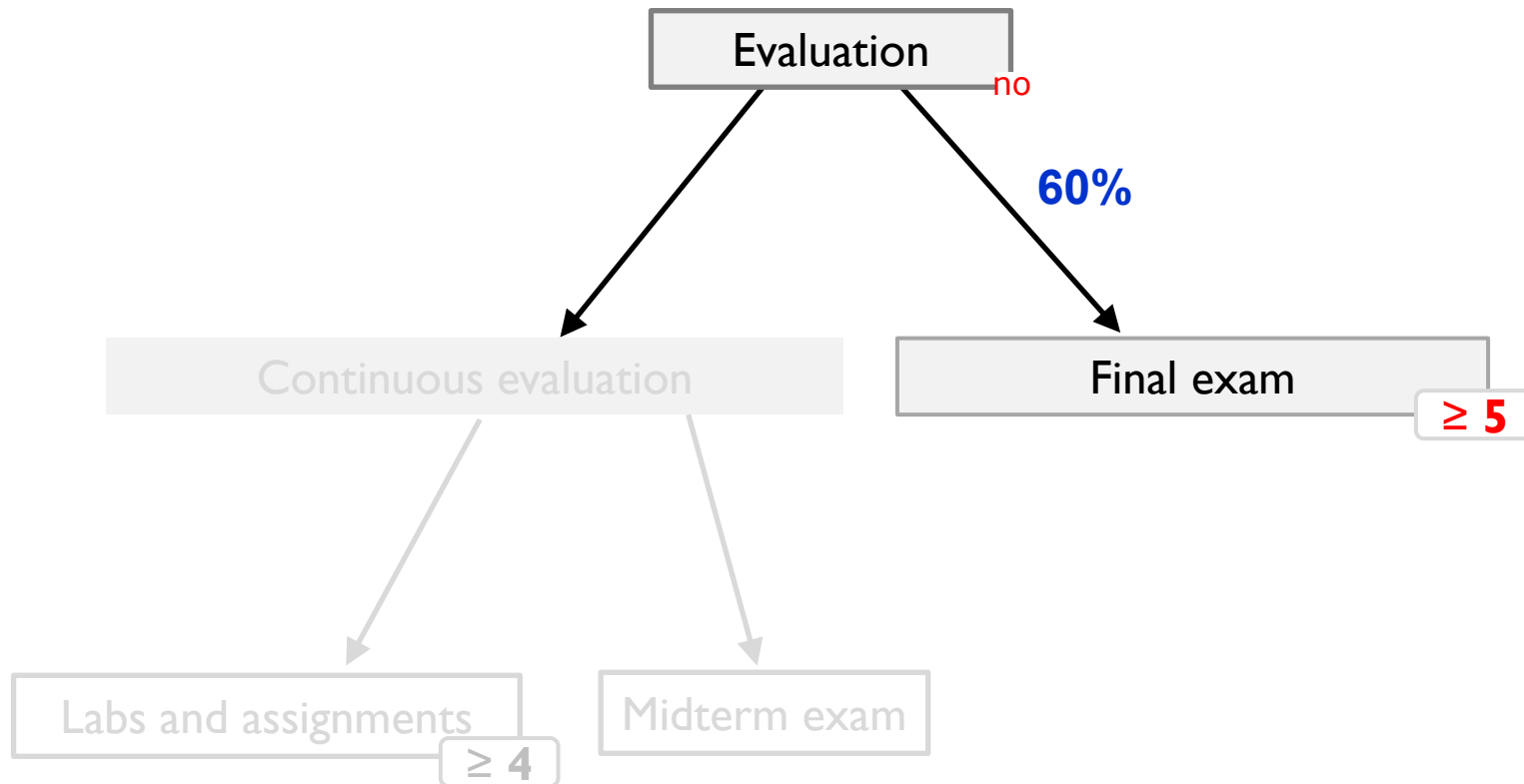
- Ordinary call + continuous eval. is followed:



Evaluation

	Ordinary call	Extraordinary call
Continuous eval. is followed	Ordinary call w/ continuous eval.	extraordinary call w/ continuous eval.
Continuous eval. is NOT followed	Ordinary call w/o continuous eval.	extraordinary call w/o continuous eval.

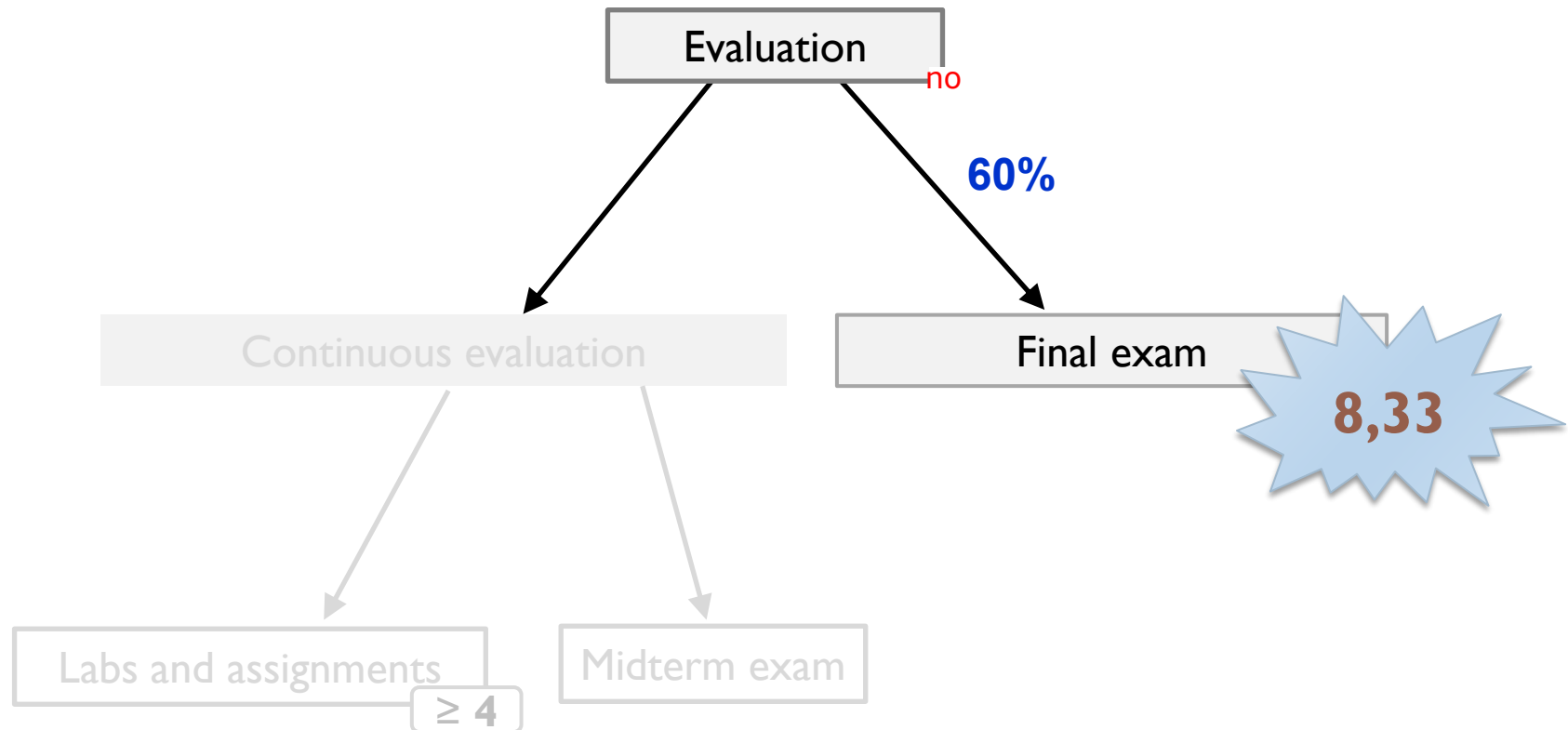
- Ordinary call + continuous eval. is not followed:



Evaluation

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Continuous eval. is followed	Ordinary call w/ continuous eval.	extraordinary call w/ continuous eval.
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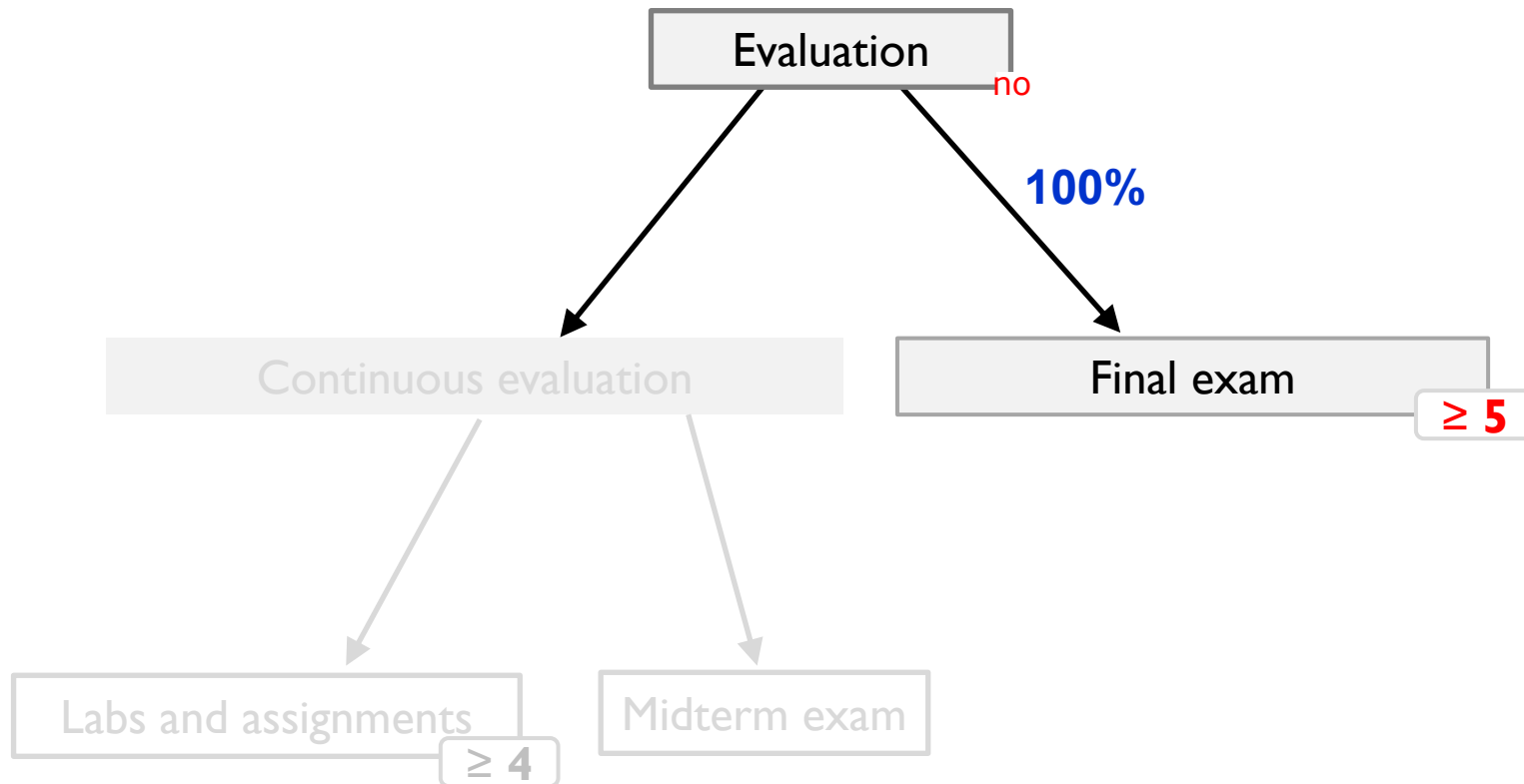
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Evaluation

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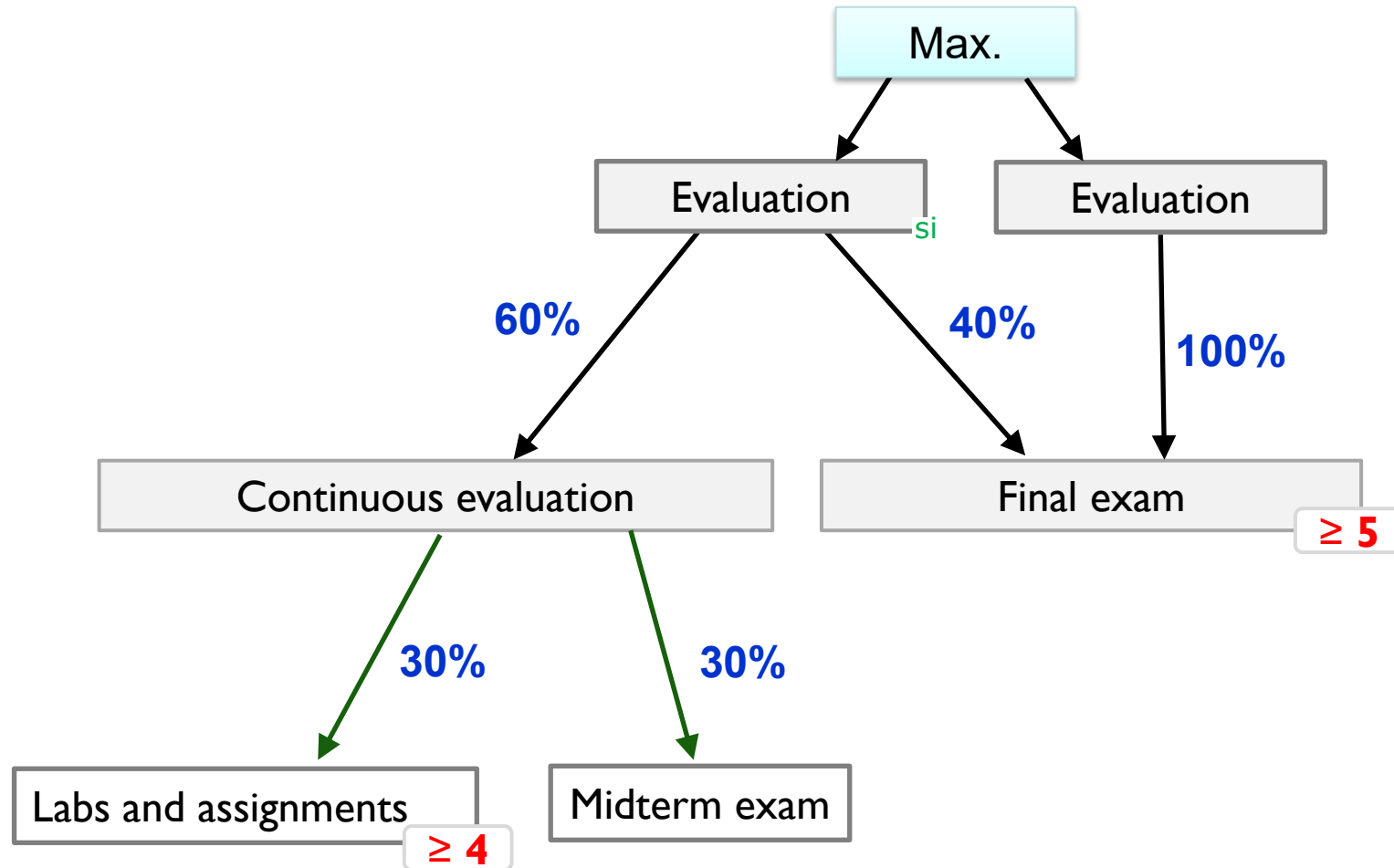
- **Extra**ordinary call + continuous eval. is not followed:



Evaluation

	Ordinary call	Extraordinary call
Continuous eval. is followed	Ordinary call w/ continuous eval.	extraordinary call w/ continuous eval.
Continuous eval. is NOT followed	Ordinary call w/o continuous eval.	extraordinary call w/o continuous eval.

- **Extra**ordinary call + continuous eval. is followed:



Continuous evaluation

Labs and assignments

- ▶ Practical and lab work weighting: 30%
 - ▶ In groups of up to **two** students (same teaching group)
 - ▶ Laboratories (10% of L&A):
 - ▶ For each of the 4 laboratories, students must submit a practical exercise that will be completed and turned in during the same session
 - ▶ Assignments (90% of L&A):
 - ▶ **The minimum average grade** for all practicums must be a **4**
- ▶ If plagiarism is detected in practical assignments, both parties involved (those who copied and those who were copied from) will receive a grade of 0
- ▶ The use of **AI is not permitted**. Its use to assist with or complete any assignment in this course will result in academic penalties. If its use is detected in any submitted assignment, the assignment will receive a grade of zero.

Evaluation

midterm exam

- ▶ Midterm Weight: 30%
 - ▶ To be taken **individually**
 - ▶ Midterm:
 - ▶ A midterm exam will be administered
 - ▶ Duration: ~90 minutes
 - ▶ **All** knowledge acquired by the student up to that point will be assessed
- ▶ The midterm exam will be administered in the class in which the student is enrolled on the designated day.
 - ▶ Students will not be permitted to take the midterm exam in a group other than the one in which they are enrolled.
 - ▶ The exam cannot be taken on any date other than the one announced.

Evaluation

final exam



- ▶ It includes **all** the content of the subject:
all the theoretical and practical content of the course
 - ▶ The minimum grade in the final exam will be **5**
 - ▶ If you do not take this exam, it will appear as **not presented** (even if you have passed the continuous evaluation.
- ▶ **No** reference material may be used for the exam, nor may it be copied. No electronic devices is allowed.
- ▶ It will be necessary to present the **ID card** or **university card** to take the exam.

How important is continuous evaluation

	Results most recent courses
Students following continuous assessment	78%
Students who pass the continuous evaluation	74%
Students who pass the continuous evaluation with respect to those who follow it	89%
Students who passed the course at the end but dropped out of continuous assessment	< 1 %
Students who passed the continuous assessment and have passed the course at the end	90%
Approved students	70%
Students not presented	19%
Failing students	11%

ARCOS Group

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Introduction to the course

Computer Structure

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Complementary materials

- ▶ [Computer History Museum](#)
- ▶ [Museo virtual de la Informática,
Universidad de Castilla-la Mancha](#)
- ▶ <https://www.computer.org/cms/Computer.org/Publications/timeline.pdf>
- ▶ [The EDSAC Simulator](#)
- ▶ [IBM archives](#)
- ▶ [Charles Babbage Institute](#)
- ▶ [Museo histórico de la Informática,
Universidad Politécnica de Madrid](#)